



Web Visualization Techniques and Resources for the GeoZarr Specification

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1 Get set up

What you'll learn today



WHAT YOU'LL LEARN TODAY

Get Set up

- <https://github.com/EOPF-Explorer/foss4g-eu-26-workshop>

What you'll learn today

- Read cloud-native GeoZarr straight from S3
- Style and combine bands in the browser
- Render server-side tiles and band-math (NDVI, SWIR) with TiTiler
- Build a scrollytelling story
- Build a STAC-driven dashboard
- Visualize Interactive components in Jupyter notebooks

2 Introduction to EOPF

Why cloud-native



WHAT IS EOPF?

Earth Observation Processing Framework

- ESA's initiative for modernised Copernicus data
- A new approach to processing and distributing Sentinel data

Standardised Format Across Missions

- Single meta-model applicable to all Sentinel-1, -2, and -3 (land) missions
- Consistent structure regardless of which satellite captured the data

EOPF PRODUCT

MEASUREMENTS

QUALITY

CONDITIONS

ATTRIBUTES

CLOUD-NATIVE: THE BENEFITS



Traditional Workflow

- Download entire datasets (often 500MB to several GB per scene)
- Store locally before analysis
- Limited by storage capacity and bandwidth



Cloud-Native Workflow

- Dynamic rendering without downloads
- Access only the data you need, when you need it
- Work directly with data in the cloud and scale up your analysis

WHAT IS ZARR?

A Cloud-Optimised Data Format

- Multi-dimensional array storage format
- Designed specifically for cloud environments
- Open specification with growing ecosystem

Key Technical Features



Chunking:

Data divided into manageable pieces



Partial retrieval:

Access only what you need



Parallel access:

Multiple chunks can be read simultaneously



Compression:

Efficient storage without sacrificing access speed

RESOURCES FOR THE TRANSITION TO EOPF

TOOLKIT



SENTINEL ZARR SAMPLES



[LEARN THE BASICS](#)

[DISCOVER THE DATA](#)

[VISUALISE: TELL A STORY](#)

EXPLORER



3 What is GeoZarr

What does it solve? and how?



GEOZARR SPECIFICATION

GeoZarr Specification

OGC Geospatial Zarr Standard

Multiscales

Encoding pyramid information for
multiresolution data

Projection

CRS and datum encoding for
geospatial data

Spatial

Array indices to spatial coordinates
relationship

Zarr v3 Core



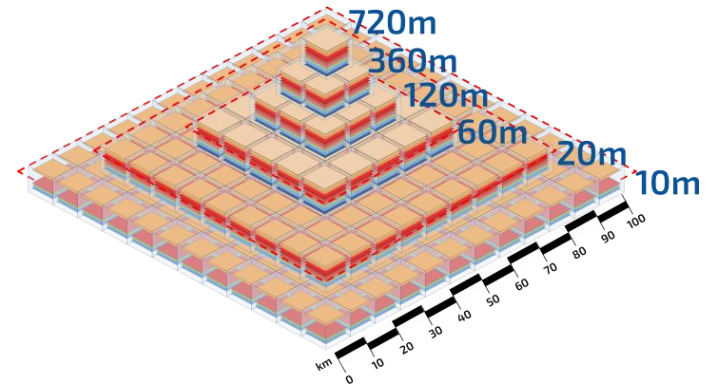
MULTISCALES ATTRIBUTE EXTENSION

The problem: one full-resolution band is huge

- One 10 m band is $10980 \times 10980 \approx 120 \text{ MP} \approx 480 \text{ MB}$

GeoZarr solves it with an overview pyramid

- This scene: six levels
- $r10m \rightarrow r20m \rightarrow r60m \rightarrow r120m \rightarrow r360m \rightarrow r720m$
- $r60m$ is 1830×1830 ; $r720m$ just 152×152



SPATIAL CONVENTION



b04[y=2, x=4]

x = 600270 m

y = 7499850 m

1-D x / y coordinate arrays stored beside the bands

The problem: which axis is which, and where's this pixel?

GeoZarr solves it via named axes and coordinate arrays.

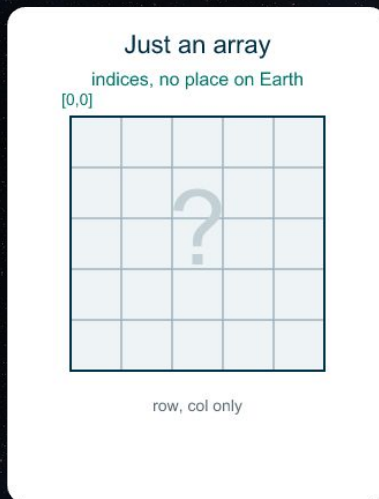
GEOSPATIAL PROJECTION

ATTRIBUTE EXTENSION

The problem: How to position on an Earth map?

- Indices are row, col
- You can't overlay, reproject, or measure

GeoZarr solves it by CRS and an affine transform.



+ CRS + affine transform

```
proj.code = "EPSG:32625"  
grid_mapping -> spatial_ref  
  
x = 600000 + col*10  
y = 7500000 - row*10
```



4 Data Access

EOPF Explorer STAC API

FIND A GEOZARR IN THE EOPF EXPLORER STAC

Browse the catalogue

- STAC browser: api.explorer.eopf.copernicus.eu/browser
- API: api.explorer.eopf.copernicus.eu/stac
- Collection: sentinel-2-l2a

Filter down to exactly one scene.

- bbox + datetime + eo:cloud_cover → the newest matching item
- One item = one acquisition = one .zarr store

The screenshot displays the EOPF Sentinel Explorer STAC API interface. At the top, there's a navigation bar with 'Browse', 'Search', and 'Language: English'. The main heading is 'Sentinel-2 Level-2A'. Below this, a 'Description' section explains the product: 'The Sentinel-2 Level-2A Collection 1 product provides orthorectified Surface Reflectance (Bottom-Of-Atmosphere: BOA), with sub-pixel multispectral and multitemporal registration accuracy. Scene Classification (including Clouds and Cloud Shadows), AOT (Aerosol Optical Thickness) and WV (Water Vapour) maps are included in the product.' A 'License' section mentions 'Legal notice on the use of Copernicus Sentinel Data and Service Information' and 'Temporal Extent: 2015-01-01 00:00:00 UTC-2026-12-31 14:09:01 UTC'. A map shows the geographical area with a highlighted region. To the right, a 'Providers' section lists 'European Commission', 'ESA', and 'EOPF Sentinel Zarr Samples Service'. The 'Metadata' section is partially visible. On the far right, a grid of 'Items' is shown, each with a thumbnail and a 'Zarr' button. The first item is 'S2A_MSIL2A_20260528T103701_N0512_R008_T31TGT_20260528T191020' with a 'Zarr' button and a timestamp of '2026-05-28 10:37:01 UTC'. The second item is 'S2A_MSIL2A_20260627T104041_N0512_R008_T31TGT_20260627T183908' with a 'Zarr' button and a timestamp of '2026-06-27 10:40:41 UTC'. The third item is 'S2B_MSIL2A_20260531T03619_N0512_R008_T31TGT_20260531T143638' with a 'Zarr' button and a timestamp of '2026-05-31 03:61:9 UTC'. The fourth item is 'S2B_MSIL2A_20260531T03619_N0512_R008_T31TGT_20260531T143638' with a 'Zarr' button and a timestamp of '2026-05-31 03:61:9 UTC'.

INSIDE THE ITEM - ASSETS

STAC item > assets

```
"assets": {  
  "reflectance": {  
    "href": ".../measurements/reflectance"  
    "type": "application/vnd.zarr; version=3; profile=multiscales"  
    "roles": ["data", "reflectance"]  
  },  
  "SCL_20m": ".../conditions/mask/l2a_classification/r20m/scl"  
  "AOT_10m": ".../quality/atmosphere/r10m/aot"  
  "WVP_10m": ".../quality/atmosphere/r10m/wvp"  
}
```

The GeoZarr reflectance group
media type carries the multiscales profile
href goes straight into eo3-map

Other groups in the same store
SCL — scene classification
AOT · WVP — atmosphere

INSIDE THE ITEM - LINKS

STAC item › links

```
"links": [  
  { "rel": "store", "href": "https://s3.../S2A_..._T28WDD_...zarr" }  
  { "rel": "xyz", "href": ".../tiles/WebMercatorQuad/{z}/{x}/{y}.png"  
    ?variables=/measurements/reflectance:b04 ...b03 ...b02 }  
  { "rel": "tilejson", "href": ".../WebMercatorQuad/tilejson.json" }  
  { "rel": "viewer", "href": ".../items/.../viewer" }  
  { "rel": "via", "href": "explorer.eopf.copernicus.eu/.../items/..." }  
]
```

Root .zarr store
open it directly with zarrita.js

TiTiler XYZ tiles
{z}/{x}/{y}.png, bands pre-combined
drop into any web map — no client GPU